

A Cubic Gate-Opening Law for Wide Binaries: a Zero-Parameter Prediction in the Separation Window Current Analyses Exclude

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Abstract

Wide binaries at separations of a few kAU are the one regime where the weak-field force law is probed below the MOND acceleration scale a_0 in a system containing negligible dark matter under any model, so a collisionless-halo picture has essentially no free parameter with which to respond. Published analyses of the Gaia catalogues nonetheless disagree: Chae (2023–2025) reports a boost, while Banik, Pittordis & Sutherland (2024) report a Newtonian null, and the two differ by $\Delta\gamma_v \approx 0.174$ — about $2.1\times$ the entire boost predicted by a de Sitter–Unruh modified-inertia (MI) framework with $a_0 = cH_\Lambda/Z$, $Z = \sqrt{(32\pi/3)}$. An amplitude test cannot survive that spread. We point out that the same framework, when its committed low-pass response gate $G(\omega) = (1 + i\omega/\omega_c)^{-1}$ with $\omega_c \in [1.782, 2.211] \times 10^{-14} \text{ s}^{-1}$ is applied to the *orbital* frequency of the pair, makes a much sharper prediction along a *shape* axis that has never been examined. A binary’s orbital frequency $\Omega = (GM/s^3)^{1/2}$ exceeds ω_c throughout the analysed separation range, so the gate is shut and the framework predicts a Newtonian result at 2–30 kAU. Because Ω falls as $s^{-(3/2)}$, the gate does not open at a threshold but as $\mathbf{Re\,G} \rightarrow (\omega_c^2/GM)\cdot s^3$, giving $\gamma_v(s) - 1 \simeq \frac{1}{2}\mathbf{v}(\mathbf{y_ext}) - 1s^3$ with the exponent equal to 3 by derivation rather than by fit and the amplitude fixed entirely by ω_c , the measured pair mass, and the measured Galactic external field. There are no free parameters and the exponent is not adjustable. Three distinct mass powers follow from the same closed form and are recovered numerically: the knee radius $r_{\text{gate}} = (GM/\omega_c^2)^{1/3} \propto M^{0.3333}$, the external-field plateau radius $r_{\text{efe}} = (GM/g_{\text{ext}})^{1/2} \propto M^{0.5000}$, and the fixed-separation excess $\propto M^{-0.9935}$. Every published contaminant channel — chance alignment, unresolved tertiaries, close-binary contamination — carries a fixed velocity scale and therefore rises as $s^{1/2}$; exponent 0.5 against 3.0. The observational consequence is the point of this note: γ_v rises from 1.0125 at 30 kAU, the cut adopted by every published analysis, to 1.0932 at 200 kAU, within the reach of the public El-Badry, Rix & Heintz (2021) catalogue — a factor of 7 in excess, across a window excluded by a cut chosen to control contamination, for reasons unrelated to this prediction. At the 30 kAU cut the predicted excess (1.2×10^{-2}) lies an order of magnitude below the inter-group systematic, so the conventional window *cannot* test the framework; that is itself the result. **We state the costs plainly.** The prediction is conditional on an unresolved question about the framework’s own kernel: whether $K(\square_u)$ responds to the acceleration *magnitude* $|a|$, which is constant on a circular orbit and would leave the gate open (recovering the ungated $\gamma_v \approx 1.09$), or acts as the linear operator its covariant form suggests, which selects the gated branch

computed here. The two branches lie $4.9\text{--}5.8\sigma$ apart in the framework's own DR4 error model and the choice must be frozen before Gaia DR4. Feasibility is also unresolved: contamination grows with separation too, and whether signal-to-contamination *improves* beyond 50 kAU cannot be settled without the catalogue in hand. Finally, a confirmed s^3 rise would establish a *frequency* scale ω_c , not the coefficient Z . The value $Z = 5.7888$ and the conventional $2\pi = 6.2832$ differ by 7.87%, and no arena examined here separates them. A confirmed rise would also constrain the weak-field force law, not the matter content. a_0 's value, Z , the response sign, and ω_c remain postulated.

1. The framework, and the credit it owes

The kernel used throughout is the framework's own de Sitter–Unruh interpolation

$$\nu(y) = \sqrt{1 + 1/y}, \quad y = g_{\text{bar}}/a_0, \quad g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}},$$

with the acceleration scale tied to the cosmological constant,

$$a_0 = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}}, \quad Z = \sqrt{32\pi/3} = 5.78881.$$

This kernel is not new and the identity is not new. Both are Milgrom (1999, *Phys. Lett. A* **253**, 273), Eqs. (8)–(9), derived from the de Sitter–Unruh temperature difference, with coefficient $2cH_\Lambda$ — a ratio of exactly $2Z$ to the value used here. Milgrom (1999, 2015) and Smolin (2017, *Phys. Rev. D* **96**, 083523) had already tied a_0 to Λ with a 2π coefficient. The distinctive content of the present framework is the *coefficient* cH_Λ/Z together with a covariant modified-inertia completion; priority for tying a_0 to Λ is not claimed, and cannot be, since Z and 2π differ by only 7.87%.

Numerically, $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ on the canonical (pure- Λ) footing and $1.13 \times 10^{-10} \text{ m s}^{-2}$ on the alternative cH_0/Z footing. **Both are carried through every number below**; they differ by $\approx 21\%$ and no result here depends on the choice.

The framework's modified-inertia realisation carries a low-pass response gate

$$G(\omega) = \frac{1}{1 + i\omega/\omega_c}, \quad \text{Re } G = |G|^2 = \frac{1}{1 + (\omega/\omega_c)^2},$$

with $\omega_c \in [1.782, 2.211] \times 10^{-14} \text{ s}^{-1}$ on the canonical footing. The lower edge is forced by the galactic rotation-curve fits; the upper edge by lunar laser ranging. **ω_c is anchored by no independent theoretical argument** — a dimensional census of the framework's intrinsic scales places every one of them 3.2–5.6 decades away — and it is the framework's most exposed quantity.

2. Why the gate is shut in the analysed window

A binary of total mass M at separation s presents an orbital frequency $\Omega = (GM/s^3)^{(1/2)}$. For $M = 1.5 M_\odot$ at $s = 10 \text{ kAU}$,

$$\Omega = 2.44 \times 10^{-13} \text{ s}^{-1} = 11.0 \times \omega_c^{(\text{upper})},$$

i.e. above the *entire* committed window. The gate is therefore shut, $\text{Re } G = 0.005\text{--}0.008$, and the predicted velocity boost collapses to $\gamma_v = 1.0004\text{--}1.0006$ — within 0.04σ of Newton in the framework's own DR4 error model ($\sigma_\gamma = 0.0191$ at $N = 30\,000$).

Two radii govern the pair, and they run in opposite directions with separation:

$$r_{\text{efe}} = \sqrt{GM/g_{\text{ext}}} \quad (\text{external field takes over}), \quad r_{\text{gate}} = (GM/\omega_c^2)^{1/3} \quad (\Omega = \omega_c).$$

Since $\Omega \propto s^{(-3/2)}$ while $g \propto s^{(-2)}$, one has $r_{\text{gate}} > r_{\text{efe}}$ generically. For $1.5\,M_\odot$ with $g_{\text{ext}} = V^2/R = 1.726 \times 10^{-10} \text{ m s}^{-2}$ ($V = 233 \text{ km s}^{-1}$, $R = 8.2 \text{ kpc}$): $r_{\text{efe}} = 7.18 \text{ kAU}$ and $r_{\text{gate}} = 49.5\text{--}57.3 \text{ kAU}$. **Between them the pair is already below a_0 but still gate-shut, and must look Newtonian.**

This does not damage the galactic case, which is the check that makes the gated branch admissible at all. Galactic orbital frequencies are $\Omega = v/r$, some nine orders smaller. Across the Milky Way solar circle and outer disk, dwarfs, low-surface-brightness disks, massive spirals, and the inner disk, the largest value found is $\Omega/\omega_c = 0.364$, giving $\text{Re } G \geq 0.883$. The gate is open in galaxies and shut in wide binaries: the radial acceleration relation and the baryonic Tully–Fisher relation are untouched. The gate makes wide binaries a *different regime*, not the same physics at smaller scale.

3. The cubic law

Deep in the gate-shut regime $\Omega \gg \omega_c$, so $\text{Re } G \rightarrow (\omega_c/\Omega)^2$, and substituting $\Omega = (GM/s^3)^{(1/2)}$,

$$\boxed{\text{Re } G \rightarrow \frac{\omega_c^2}{GM} s^3, \quad \gamma_v(s) - 1 \simeq \frac{1}{2} \left[\nu(y_{\text{ext}}) - 1 \right] \frac{\omega_c^{(2)}}{GM} s^{(3)}}$$

saturating at $\frac{1}{2}[\nu(y_{\text{ext}}) - 1]$ once $\Omega < \omega_c$. The local logarithmic slope $d \ln(\gamma_v - 1)/d \ln s$, evaluated numerically, is **3.00** at 5 kAU and falls monotonically through the knee to 0.06 once saturated.

The exponent is **derived, not fitted**, and is not adjustable. The amplitude contains ω_c (committed), M (measured per pair), and $y_{\text{ext}} = g_{\text{ext}}/a_0$ (measured Galactic field, postulated a_0). **There are no free parameters.**

Three mass powers follow from the same closed form, and are recovered numerically over $0.5\text{--}5.0\,M_\odot$:

quantity	measured	analytic
r_{gate}	$M^{(0.3333)}$	$M^{(1/3)}$
r_{efe}	$M^{(0.5000)}$	$M^{(1/2)}$
excess at fixed $s = 10 \text{ kAU}$	$M^{(-0.9935)}$	$M^{(-1)}$

The third is **asymptotic**, valid only where $\text{Re } G \ll 1$. Measured at 30 kAU it is $M^{(-0.857)}$, not $M^{(-1)}$, because low-mass pairs are by then near their own knee (r_{gate} shrinks as $M^{(1/3)}$). Any

fit must use the local slope rather than assume a global $1/M$. We report both values rather than the favourable one.

4. The discriminant

Every published contaminant channel carries a *fixed velocity scale* σ_v and therefore produces a scaled relative velocity rising as $s^{(1/2)}$: chance alignment (El-Badry, Rix & Heintz 2021, via $R_{\text{chance_align}}$), unresolved tertiaries and hierarchical systems (Peñarrubia 2021; Tyler, Green & Goodwin 2023), and close-binary contamination. **Exponent 0.5 against 3.0** — a factor of six in logarithmic slope. Contamination and signal therefore have *different* separation dependences, which makes a joint fit for both well posed in principle. This is strictly stronger than the amplitude test, which uses no shape information at all and is hostage to a 0.174 inter-group systematic amounting to $2.1\times$ the entire ungated signal.

5. The window nobody has examined

Because the excess grows as s^3 , essentially the whole signal lies at large separation ($M = 1.5 M_\odot$, canonical footing, lower ω_c edge):

s [kAU]	γ_v	note
30	1.0125	the cut used by every published analysis
50	1.0392	—
100	1.0808	—
200	1.0932	the public catalogue’s actual reach

The predicted excess grows by a **factor of 7** from the conventional cut to the catalogue limit. The El-Badry, Rix & Heintz (2021, *MNRAS* **506**, 2269) Gaia DR3 catalogue extends to $\approx 1 \text{ pc} = 206 \text{ kAU}$ and is public; the data are already taken. Analyses cut near 30 kAU because chance-alignment contamination rises with separation — a motivation entirely unrelated to this prediction.

At the 30 kAU cut the predicted excess is 1.2×10^{-2} , an order of magnitude below the demonstrated inter-group systematic of 0.174. **The conventional window cannot test this framework.** That is the central observational claim of this note.

6. What is not claimed

- **The branch is favoured but not closed, and this whole result is conditional on it.** $K(\square_u)$ is a *linear* operator, while $|a| = (a^\mu a_\mu)^{(1/2)}$ is not a linear functional of the trajectory — a linear kernel cannot by itself sense the acceleration magnitude that $v(y)$ depends on. That obstruction points to the gated (AC) branch. **The framework’s own field-theory analysis points the same way:** in the matter sector the kernel reduces to $K(\square_u) \rightarrow K(-q_0^2)$, *a function of the frequency q_0 alone*, which is precisely the AC reading used here. On the ungated (DC) branch — a kernel responding to $|a|$, constant on a circular orbit — this entire note is void and $\gamma_v \approx 1.09$ stands instead. The branches lie $4.9\text{--}5.8\sigma$ apart in the framework’s own DR4 error model. **The choice must be frozen before DR4 or the test is post hoc.** The same fork governs the framework’s apparent $\dot{G}/G$ floor, since the lunar mean motion exceeds ω_c by $\approx 1.5 \times 10^8$.

- **A deeper question sits underneath this one, and it is not addressed here.** If the kernel is a function of frequency (or, in the eikonal limit of a field, of momentum) rather than of acceleration, it is a separate and prior question whether the construction reproduces an acceleration-dependent inertia law at all — i.e. whether $v(y)$ with $y = g/a_0$ is recovered, as opposed to a momentum-dependent response. The framework’s own notes raise this explicitly and do not settle it. The present note assumes the MI law as given and asks only what the gate does to it in a wide binary; it is not evidence on that prior question, and a reader should not take it as such.
- **Feasibility beyond 50 kAU is unresolved and may be fatal to a DR3 test.** Contamination also grows with separation. Whether signal-to-contamination *improves* beyond 50 kAU is an empirical question that cannot be answered without the catalogue in hand. If it does not, this prediction waits for DR4’s astrometry and higher N rather than being testable now. We state this rather than assume the favourable case.
- **A confirmed s^3 rise would not test Z .** It would establish a frequency scale ω_c . The coefficient $Z = 5.7888$ and the conventional $2\pi = 6.2832$ differ by 7.87%, both lie inside the $\pm 16\%$ empirical a_0 box, and no arena examined here separates them. Separately and against interest: matching Chae (2024b)’s central γ on this framework’s own kernel would require $a_0 \approx 1.9\times$ the canonical value, which at face value disfavors the Λ -tied coefficient specifically.
- **Nothing here bears on whether dark matter exists.** A wide-binary result constrains the weak-field *force law*, not the matter content. $\gamma_v = 1$ is Newton’s prediction, shared by Λ CDM, by this framework’s own gated branch, and by hybrid modified-force-plus-particle models.
- **a_0 ’s value, Z , the response sign, and ω_c are postulated, not derived.**

7. Falsification

The prediction fails if, in a contamination-controlled sample beyond ≈ 50 kAU, the scaled relative velocity excess is (i) absent, (ii) present with a logarithmic slope inconsistent with 3 in the gate-shut regime, (iii) present but without the knee moving as $M^{1/3}$, or (iv) present with an amplitude inconsistent with ω_c^2 / GM on both footings and both window edges. Any of these kills the gated branch. A Newtonian result *inside* 30 kAU falsifies nothing, on either branch — which is precisely the defect in the current pre-registration that this note exists to correct.

8. Reproducibility

Every number is printed by two committed scripts, both exit 0, with no hard-coded verdicts and both footings and both ω_c edges carried throughout:

- `real_research/reviews/mi_wb_gate_fork_2026.py` — the DC/AC fork, the gate-shut computation, the galactic-safety check, and the dead zone (11/11 checks).
- `real_research/reviews/mi_wb_cubic_rise_2026.py` — the s^3 law, the slope measurement, the three mass exponents, the profile table, and the feasibility caveat (10/10 checks).

Two assertions failed their own checks on first run and were corrected in place rather than patched: an overstated 200-vs-30 kAU excess ratio (claimed $>10\times$, actually $7\times$) and an overstated global $1/M$ scaling (actually $M^{(-0.857)}$ at 30 kAU, asymptotic only). Both corrections are recorded in the scripts and in the commit history.

References

Banik, I., Pittordis, C., Sutherland, W., et al. 2024, *MNRAS* — wide-binary Newtonian null. Bekenstein, J. D., & Milgrom, M. 1984, *ApJ* **286**, 7. Chae, K.-H. 2023–2025, *ApJ* — wide-binary boost claims. El-Badry, K., Rix, H.-W., & Heintz, T. M. 2021, *MNRAS* **506**, 2269 — the public Gaia DR3 catalogue. Milgrom, M. 1983, *ApJ* **270**, 365. Milgrom, M. 1999, *Phys. Lett. A* **253**, 273 — Eqs. (8)–(9): the kernel and identity used here. Milgrom, M. 2015, *Can. J. Phys.* **93**, 107. Peñarrubia, J. 2021 — wide-binary contamination. Smolin, L. 2017, *Phys. Rev. D* **96**, 083523. Tyler, J., Green, A., & Goodwin, S. 2023 — hierarchical contamination.